ESERCIZI

es.1)

- $f(x) = \dfrac{1}{1+x}$

$\text{Dom}(f) = (-\infty, -1) \cup (-1, +\infty)$

- $\lim\limits_{x \to +\infty} f(x) = \dfrac{1}{1+\infty} = \dfrac{1}{\infty} = \varnothing$

- $\lim\limits_{x \to -\infty} f(x) = \dfrac{1}{1-\infty} = \dfrac{1}{-\infty} = -\dfrac{1}{\infty} = \varnothing$

- $\lim\limits_{x \to -1^-} f(x) = \dfrac{1}{1-1^-} = \dfrac{1}{\varnothing^-} = \infty$

SIAMO LEGGERMENTE A SINISTRA

- $\lim\limits_{x \to -1^+} f(x) = \dfrac{1}{1-1^+} = \dfrac{1}{\varnothing^+} = \infty$

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$\dfrac{1}{\varnothing} = \infty \notin \mathbb{R}$

$\dfrac{1}{\infty_{\notin \mathbb{R}}} = \varnothing$

es.2)

$f(x) = \dfrac{x}{\underbrace{x^2+1}_{\geq 1 \ \forall x \in \mathbb{R}}}$

- $\lim\limits_{x \to +\infty} f(x) = \dfrac{\infty}{\infty^2+1} = \dfrac{\cancel{\infty}}{\infty^{\cancel{2}}} = \dfrac{1}{\infty}$

es. 9)

$$f(x) = \frac{\sqrt{1+x} - 1}{x^2} \qquad Dom(f) = [-1, 0) \cup (0, +\infty)$$

$$\bullet \lim_{x \to 0^+} \frac{\sqrt{1+x} - 1}{x^2} = \frac{\sqrt{1+0^+} - 1}{(0^+)^2} = \frac{0}{0} = \text{FORMA INDETERMINATA}$$

$$\bullet \lim_{x \to 0^+} \frac{(\sqrt{1+x} - 1)}{x^2} \cdot \frac{(\sqrt{1+x} - 1)}{\sqrt{1+x} - 1} = \lim_{x \to 0^+} \frac{\cancel{1} + x \ \ - \cancel{1}}{x^2(\sqrt{1+x} - 1)} =$$

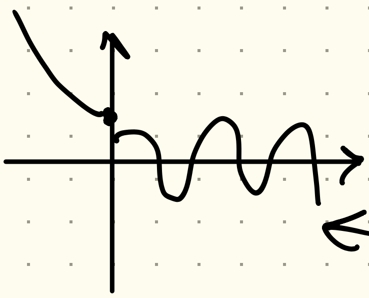
$$= \lim_{x \to 0^+} \frac{+x}{x^2(\sqrt{1+x} + 1)} = \lim_{x \to 0^+} \frac{1}{x^2(\sqrt{1+x} + 1)} = \frac{1}{0 \cdot 2} =$$

$$= \frac{1}{2} \cdot \left(\frac{1}{0^+}\right) = +\infty$$

$$\bullet\ g(x) = \overset{a\ -\ b}{\sqrt{1-x} - 1} \cdot \frac{\overset{a+b}{\sqrt{1+x} + 1}}{\sqrt{1+x} + 1} = \frac{\overset{a^2}{(1+x)} - \overset{b^2}{1}}{\sqrt{1+x} + 1} = \frac{x}{\sqrt{1+x} + 1}$$

es)

$$f(x) = \begin{cases} \dfrac{\sin(x)}{x} & x > 0 \\ x^2 + a & x \leq 0 \end{cases}$$



$$\frac{\sin(x)}{x}$$

- $\lim_{x \to 0^+} \frac{\sin(x)}{x} = 1^-$

$$f(0) = (x^2 + a)\Big|_{x=0} = 0^2 + a = a$$

es. 5)

$$f(x) = \begin{cases} \dfrac{e^{-\frac{1}{x}}}{x^a} & x < a \\ a\,x & x \geq 0 \end{cases}$$

$f(0) =$